NYANZA ADVENTIST SECONDARY SCHOOL

FORM FOUR PREMOCK 1 EXAMINATIONS

BIOLOGY 1

TIME: 2:30 Hrs

Year: 2025

Instructions

1. This paper consists of three sections A, B and C with a total of **eleven (11)** questions.
2. Answer all questions in sections A and B and two (2) questions from section C.
3. Except for diagrams that must be drawn in pencil all writings should be in blue or black ink.
4. Cellular phones and any unauthorized materials are not allowed in the Examination Room.
5. Write your Examination Number on every page of your answer booklet(s)

SECTION A (16 Marks)

Answer **all** questions in this section.

1. For each of the items (i – x), choose the correct answer from among the given alternatives and write its letter in the answer booklet (s) provided.

   i) Mr Sango being a street vender, he got an accident that caused bleeding through the cuts when he tried to slash the long grass to Mr John's house. As Mr John what would you do to help before taking him to hospital?
      A. Lay Mr Sango on his back while legs raise.
      B. Dress the Mr Sango's wound and bandage it firmly.
      C. Give Mr Sango a lot of fruit juice to restore mineral lost.
      D. Induce vomiting to Mr Sango.
      E. Call professional Doctors to help Mr sango.

   ii) A woman with blood group A claims that a man with blood group B is the father of her child. A blood test reveals that the child's blood is group O. Identify the genotype of this couple that proves the alleged man cannot be excluded as a possible biological father of the child.
      A. Parent's genotype I^AI^A x I^BI^B
      B. Parent's genotype I^AI^A x I^BI^O
      C. Parent's genotype I^AI^O x I^BI^O
      D. Parents genotype I^AI^O x I^BI^B
      E. Parent's genotype I^AI^O x I^OI^B

   iii) Peter had no gloves when he was giving Juma first aid after getting head injury during interclass football league at Mkapa stadium. What else Peter could use to protect his hand?
      A. Plastic bag
      B. Plaster
      C. Piece of cloth
      D. Socks
      E. Hanker chief.

   iv) Form one students at XY secondary school were seen by their head of school collecting lizard, Grasshopper, bat, frog, and spider that they can group them to their appropriate groups. Which branch of biology concerns with their activities?
      A. Anatomy
      B. Entomology
      C. Botany
      D. Cytology
      E. Taxonomy

   v) Being a footballer Mr Kintu is likely seen to waste a lot of water through sweating. As a biologist identify the organ of the body which is responsible for the water loss.
      A. Lungs
      B. Liver
      C. Kidney
      D. Skin
      E. Pancreases

   vi) When form student poured Iodine solution accidentally into the test tube containing a food sample Y, the sample Y turned blue - black when it was mixed with iodine solution. How would you recommend for this observation?

A. Test tube contained starch
B. Test tube contained lipid
C. Test tube contained proteins
D. Test tube contained glucose
E. Test tube contained reducing sugar

vii) Ngoswe was asked by his young brother of standard six "Which structure of the leaf that allows oxygen and carbon dioxide to diffuse into and out of the leaf during gaseous exchange?" Respond to this pupil for Ngoswe.
A. Mesophyll
B. Cuticle
C. Epidermis
D. Stomata
E. Guard cells

viii) A form four student wants to prepare microscope slides showing different stages of mitosis. As a botanist what would you advise this student to use in his experiment to observe different stages of mitosis?
A. Leaf
B. Root tip
C. Stem
D. Ovary
E. Flower

ix) They are considered to be very successful group of animals on earth due to their diversity in life styles, feeding methods and even movement.
A. Molluscs
B. Invertebrates
C. Echinoderms
D. Insects
E. Arachnidan

x) A cytologist discovered a new cell in a culture with a well organised nuclei, their nucleus has nuclear bound membrane. As a biologist suggest the type of this cell.
A. Eukaryote
B. Prokaryote
C. Animal cell
D. Plant cell
E. Retro cell

2. Match the process of each stage of mitosis in **LIST A** with the correct stage in **LIST B** by writing the letter of the correct response beside the item number.

LIST A	LIST B
i) Chromatids tend to shorten, thicken and become visible. ii) Chromatids uncoil, lengthen, and are not clearly seen. iii) Chromosomes line up at the equator of the spindle fibre. iv) Division of the cytoplasm to form two new cells. v) Intensive synthesis of cell organelles and the growth of the cell. vi) The centromere split and spindle fibres contract, pulling the chromatid to the opposite pole of the cell.	A. Metaphase B. Cytokinesis C. Telophase D. Meiosis E. Prophase F. Interphase G. Anaphase H. Crossing over

SECTION B (**54 marks**)

Answer **all** questions from this section.

3. Form four students from Karibuni secondary school went study tour at bakery centre where they observed the spoiled breads in a cupboard surrounded with white particles seen like clubs in structure at the top. The tour guide asked one of the students to share some concepts with her classmate based on the guidelines below:-
    a) Name of organisms found immersed on the bread.
    b) Kingdom and phylum of the organisms in (a) above.
    c) Features that guide her to assign such name in (a) above. Give five points.

4. After teaching his student about Transportation of materials in mammals, Mr Mathias assigned his student assignment of performing surgery in mammal for better understanding of blood circulatory system. While performing surgery two blood vessels were cut by mistake whereby in one of the vessels blood was flowing out with high pressure while in the other vessel blood was flowing out with low pressure.
    a) As a biologist how would you help the student to name the two blood vessels?
    b) Assess for the variation of blood pressure in the two vessels.
    c) Identify the vessel responsible for carrying oxygenated blood and deoxygenated blood respectively.

5. It is just a human nature for the women to experience monthly discharge of the uterus lining, dead ovum and epithelial cells of the uterus as debris out of the vagina.
    a) Which biological term you would use to explain the biological phenomenon above?

b) The process in (a) above is controlled by four hormones and each hormone is performing its specific role. As biology student how would you justify the statement?
c) As an expert in food and nutrition department, which type of food would you advise the women to use for the replacement of the blood being discharged during the process in (a) above?

6. A group of student Doctors had a study visit to Maweni referral hospital at kigoma municipality where they noted a patient with running nose, dry cough, fever, loss of taste, sore throat, headache and difficult breathing.
    a) As an upcoming Doctor, what do you think could be the possible disease that the patient was suffering from?
    b) As an expert in medical laboratory share your experiences on the causes and ways of transmission of the disease in (a) above.
    c) As a recent biologist how would you control/prevent the disease in (a) above? Give four points.

7. Being a father of genetics the Australian monk Gregory Mendel decided to use *Pissum sativum* (Pea plant) in his experiment despite the presence of other beautiful flowers and good crops.
    a) How would you define genetics?
    b) As a recent father of genetics what would you suggest for the decision of Gregory Mendel? Give five points.

8. (a) How would you tabulate the differences between egg and cassava cells in four points?
    (b) Assume your biology teacher has asked you to draw a well labelled diagram of a lion cell and show parts responsible for the following:
        i) Protect and give the cell definite shape
        ii) Allows selective movement of substances across the cell
        iii) Site for chemical reaction
        iv) A home of hereditary materials.

SECTION C (30 marks)

Answer **any** two questions from this section

9. Assume you are invited to participate on world Farmer's day in your village and you are asked to talk about "Biology for development of Agriculture and medicine, farmer's voice." Prepare and write a speech in essay form concerning the potential of biology in Agriculture and Medicine field. Give five points.

10. The medical research showed that the urinary system diseases have a wide range of severity, from minor to life threatening. As a Medical Doctor how would you educate people in your country in two points on how they can escape from suffering the common urinary diseases below: -
    i) Urinary Tract Infections (UTIs)
    ii) Uraemia
    iii) Gout
    iv) Diabetes inspidus
    v) Kidney stone (renal calculi)

11. Form four biology students went Temeke hospital to collect data for their project titled "Why increasing in number of drugs addicted people to our society?" As a medical Doctor of Temeke hospital how would you respond to these students with six points?